

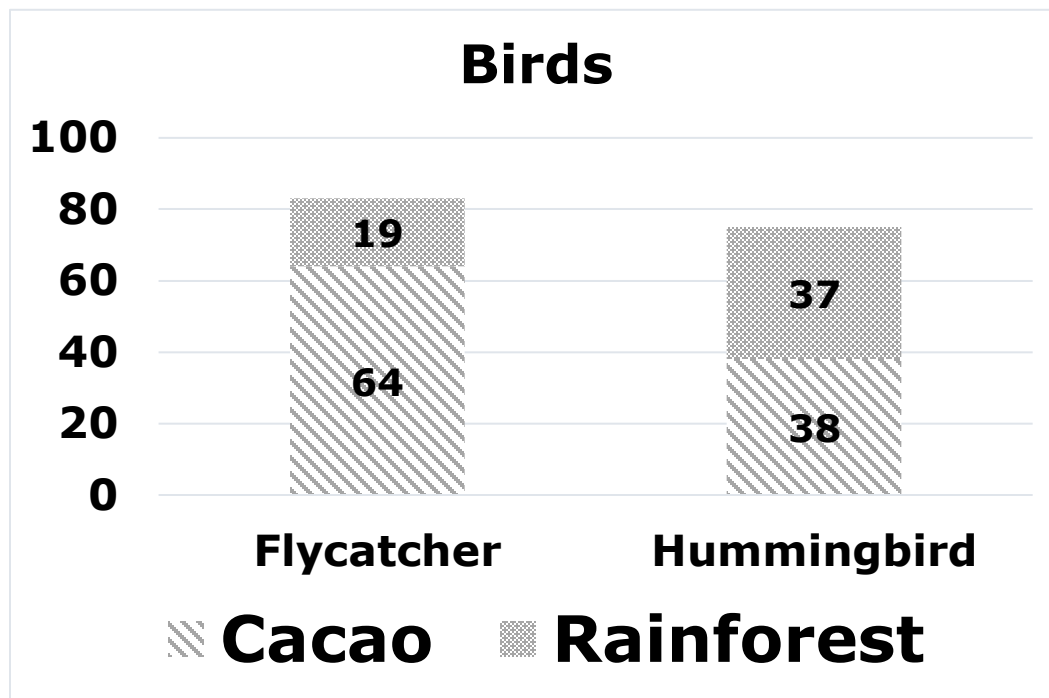
1. In 2021, a research group surveyed 6,423 adults to determine how they watch television. Participants choose between cable and satellite services or other services such as streaming.

An incomplete two-way frequency table is shown.

	Cable and Satellite Services	Other Services	Total
Men	1,725		3,248
Women		1,764	
Total	3,136		

- a. Determine the number of women who were surveyed.
3,175
- b. Determine the number of people who watch television using other services.
3,287
- c. Determine the joint frequency for women who watch television using cable and satellite services.
1,411
- d. List all of the values in the two-way frequency table that are marginal frequencies.
3,248; 3,175; 3,136; and 3,287

2. Skye is a biologist who conducted a study on the abundance of birds on cacao farms in Costa Rica compared to the abundance of birds in a nearby rainforest. Skye used a segmented bar graph, as shown, to display the data for two different species of birds.



- a. Use the segmented bar graph to complete the two-way frequency table.

	Flycatcher	Hummingbird	Total
Cacao Farm	64	38	102
Rainforest	19	37	56
Total	83	75	158

- b. Explain using the segmented bar graph or frequency table whether there is an association between a bird's location and the abundance of birds.

There is an association between the bird's location and the abundance of birds. The abundance of hummingbirds in the rainforest and on the cacao farm is approximately equal, while the abundance of flycatchers on the cacao farm is almost $3\frac{1}{2}$ times that of the abundance of flycatchers in the rainforest. The abundance of hummingbird's is not affected by location, but the abundance of flycatchers is.

3. Students in each class at Leesburg High School were asked whether they have to do chores at home and whether they receive an allowance. Two-way frequency tables that show some of the results of the survey are shown.

a. Complete each two-way frequency table.

Freshmen				Sophomores			
	Do Get An Allowance	Do Not Get An Allowance	Total		Do Get An Allowance	Do Not Get An Allowance	Total
Do Chores	96	25	121	Do Chores	131	62	193
Do Not Do Chores	60	121	181	Do Not Do Chores	18	58	76
Total	156	146	302	Total	149	120	269
Juniors				Seniors			
	Do Get An Allowance	Do Not Get An Allowance	Total		Do Get An Allowance	Do Not Get An Allowance	Total
Do Chores	101	72	173	Do Chores	105	84	189
Do Not Do Chores	71	23	94	Do Not Do Chores	21	42	63
Total	172	95	267	Total	126	126	252

- b. Use the frequency tables for each class to create a frequency table that shows the results of the survey for all of the students at Leesburg High School who responded to the survey.

	Do Get an Allowance	Do Not Get an Allowance	Total
Do Chores	433	243	676
Do Not Do Chores	170	244	414
Total	603	487	1,090

4. A survey found that of 2,000 high school seniors surveyed 1,800 have their driver's license. Of those who have their driver's license, 1,350 have a job. Of the seniors who do not have a driver's license, 40 have a job.

- a. Complete the frequency table.

	Has a Job	Does Not Have a Job	Total
Has Driver's License	1,350	450	1,800
Does Not Have a Driver's License	40	160	200
Total	1390	610	2,000

- b. Explain how the information in the table can be used to show that there is an association between having a job and having a driver's license.

About $\frac{3}{4}$ or the seniors surveyed that have a driver's license have a job, whereas only about $\frac{1}{5}$ of those who don't have a license have a job. This shows that there is an association between having a driver's license and having a job.